Hybrid versus intrinsic Majorana nanowires, a self-consistent Hartree-Fock-Bogoliubov analysis

Francisco Lobo[1], Elsa Prada[1], Pablo San-Jose[1]
[1]*Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain*
(Dated: July 2, 2024)

We theoretically investigate and compare hybrid versus intrinsic Majorana nanowires in a self-consistent Hartree-Fock-Bogoliubov theory. The mean-field self-energy is realized as a spin-1/2 Hubbard term in the Wannier limit through a Hartree-Fock decoupling of a quartic Hamiltonian in a Nambu-spinor scheme. This approach improves on Oreg-Lutchyn minimal model because the pairing is calculated self-consistently, rather than relying on an homogeneous proximity effect. We discuss beyond minimal models how the combined effect of spin-orbit coupling and Zeeman field tangled with the emergent self-consistent superconductivity triggers a trivial-topological phase transition, along with the role of finite length on the emergence of Majorana oscillations. We show that, in theory, Majoranas can still emerge in both hybrid and intrinsic self-consistent nanowires establishing evidence for a more experimentally grounded Majorana platform. when considering the more approach.

I. INTRODUCTION

Majorana bound states, or simply Majoranas, are non-Abelian quasiparticles [??] that emerge at the edges of one-dimensional topological superconductors [??]. Among the various proposed platforms that could support Majoranas [??], the Oreg-Lutchyn minimal model [??] has arguably received the most attention. This system relies on hybrid superconductor-semiconductor (SC-SM) heterostructures which can undergo a trivial-topological phase transition under the influence of an applied Zeeman field if the SM is characterized by a strong enough spin-orbit coupling (SOC). The simplistic proposition of this model is that one considers an homogeneous uniform pairing induced into the SM purely due to proximity effects with the SC. Having artificially set a pairing's value Δ_0 throughout the whole SM wire, one can straightforwardly induce the topological phase transition just by the tuning of the Zeeman field $V_{\rm Z}$ and chemical potential μ, occurring exactly at the critical value $V_{\rm Z} = \sqrt{\mu^2 + \Delta_0^2}$.

The pairing plays two crucial roles. For starters, it opens a gap at the outer wings of the energy dispersion $\epsilon(k)$ where the Zeeman field is unimportant. Secondly, it shapes the now smaller gap, in the proximity the symmetry point Γ, commonly known as minigap. The former role eliminates the possibility of high-momentum gapless excitations, thus leaving only the chiral states near the minigap as low energy excitation. This minigap can be shown to be effectively p-wave, and hence topologically non-trivial [??]. The latter role allows us to tune the topological phase transition, signaled by the closing and reopening of the minigap, and by the emergence of topological protected Majorana zero energy modes in the edges on the nanowire.

During the last two decades, several measurement schemes of increasing complexity have been pursued to unequivocally confirm the emergence of MBSs, including tunneling [??], Coulomb [??] and quantum-dot-assisted spectroscopies [??]. In pristine Oreg-Lutchyn nanowires, a way to identify Majoranas was proposed already in 2012 [??]. Despite such experimental efforts, interpreting experimental data on the matter proved to be extremely challenging [??], even polemic [??], especially when parsing out the complex phenomena that emerge in realistic hybrid systems such as the effects of interaction [??], disorder [??] and material inhomogeneity [??]. As the field progresses, the goal remains to establish a clear and unambiguous signature of Majoranas that can withstand the scrutiny of complex, real world systems. Given these challenges, there is growing interest in exploring alternative approaches such as intrinsic nanowires, which could inherently host p-wave type topological phases without the need for hybridization with a superconductor, thus minimizing the complexities associated with hybrid structures. Moreover, to more accurately capture interaction effects and enhance the robustness of theoretical models versus experiment, it is crucial to adopt a self-consistent approach.

To realize Majorana states in intrinsic nanowires, it is essential that the system exhibits inherent superconductivity, as this is a fundamental prerequisite. Following Oreg-Lutchyn's approach, we model the system as an s-wave trivial SC, described according to BCS theory. As an inherent mean-field theory, we consider a stochastic many-body problem where all particles interactions are averaged into an (attractive) effective interaction, allowing the formation and condensation of Cooper pairs. One then solves self-consistently for this effective interaction leading to a spatially varying pairing potential (as opposed to uniform). In particular, we do this in a fixed-point routine within a Nambu-spinor scheme, reducing the interacting Hamiltonian to an effective Bogoliubov-de-Gennes (BdG) Hamiltonian (which accounts for both normal and anomalous terms) with an exclusively onsite Hubbard self-energy term. It's the strength of this Hubbard potential U that dictates the state of the system through self-consistency.

Ultimately, the significance of this work lies in the results obtained from the self-energy calculations and the

possibility of using intrinsic nanowires, providing a more comprehensive and realistic description of experimental nanowires compared to the simplified Oreg-Lutchyn (hybrid) model.

II. NANOWIRE MODELS

First and foremost, we showcase Oreg-Lutchyn minimal Majorana model [?] as our model's groundwork. Lutchyn's minimal model consists of a finite one-dimensional nanowire in the presence of an external Zeeman field $V_{\rm Z}$ and spin-orbit coupling (SOC) effect, being described by the Hamiltonian

$$\hat{\boldsymbol{H}}_0 = \sum_{\sigma_i\sigma_j} c^\dagger_{\sigma_i}\left(-\eta\partial_x^2 - \mu + \alpha_{\rm R} i\sigma_y\partial_x + V_{\rm Z}\sigma_z\right)c_{\sigma_j}, \quad (1)$$

with $\eta = \hbar^2/2m^*$, m^* the electron's effective mass, and μ the chemical potential. The SOC effect due an externally applied transverse magnetic field of Rashba strength $\alpha_{\rm R}$ and the Zeeman splitting $V_{\rm Z} = g_J\mu_B B_x/2$ due to a in-plane magnetic field B_x, with g_J the Landé $g$yromagnetic factor of the semiconductor and μ_B Bohr's magneton. Furthermore, Lutchyn then considers a s-type superconductive (SC) proximity effect of uniform pairing Δ_0 reading as $(\Delta_0 c^\dagger_\uparrow c^\dagger_\downarrow + h.c)/2$.

The major step up to the minimal model is that we do not consider an homogeneous superconductive proximity effect term but instead self-consistently calculate for the induced pairing but also magnetization and doping. This is realized in a Hartree-Fock-Bogoliubov mean-field theory derived in detail in appendix A. The mean-field self-energy is realized as a spin-1/2 Hubbard term in the Wannier limit through a Hartree-Fock decoupling of a quartic Hamiltonian in a Nambu-spinor scheme. The obtained total onsite self-energy reads as

$$\boldsymbol{\Sigma}_{ij} = \frac{1}{2}U\left[{\rm Tr}\left([\tau_z\otimes\sigma_0]\,\tilde{\boldsymbol{\rho}}_{ii}\right)[\tau_z\otimes\sigma_0] - 2\tilde{\boldsymbol{\rho}}_{ij}\right]\delta_{ij}. \quad (2)$$

This term is comprised of both Hartree and Fock self-energies as well as the pairing; their Feynman diagrams shown in Fig.(1)(a). Within the Nambu tight-binding scheme, defining the spinor $\check{\boldsymbol{c}}_i = (c_{i\uparrow}\ c_{i\downarrow}\ c^\dagger_{i\uparrow}\ c^\dagger_{i\downarrow})$, the full Hamiltonian matrix reads explicitly as

$$\begin{aligned}\boldsymbol{H} &= (f\delta_{ij} - t\delta_{ij\pm1})\,[\tau_z\otimes\sigma_0] + V_{\rm Z}\,[\tau_z\otimes\sigma_z]\,\delta_{ij}\\ &+ \frac{\alpha_{\rm R}}{2a_0}\delta^{\pm}_{ij\pm1}\,[\tau_z\otimes i\sigma_y] + \boldsymbol{\Sigma}_{ij}\end{aligned} \quad (3)$$

with $f = 2t - \mu$ the band filling ($\mu = 0$ and $\mu = 2t$ corresponding to half and zero filling respectively), $t = \eta/a_0^2$ the hopping parameter and a_0 the real-space lattice spacing. The Kronecker delta δ_{ij} denotes the on-site terms and $\delta^{\pm}_{ij\pm1}$ the 1st neighbors hoppings (front and back), and both σ and τ are Pauli matrices operating in their respectively spin and particle-hole subspaces. The BdG symmetric object $\tilde{\boldsymbol{\rho}}_{ij}$, related to the Nambu reduced density matrix $\boldsymbol{\rho}_{ij} = \langle\check{\boldsymbol{c}}^\dagger_j\otimes\check{\boldsymbol{c}}_i\rangle$, is defined in appendix A in Eq.(8) and has its underlying spin structure concealed within the bold notation. Reiterating, in the Nambu space the significant improvement to the minimal model reads as $\Delta_0\,[\tau_y\otimes\sigma_y]\,\delta_{ij} \to \boldsymbol{\Sigma}_{ij}$.

See that the sign of the on-site Hubbard potential U dictates the system's state: if, on one hand, $U < 0$ then the correlation $\rho^{\rm eh}_{i\uparrow j\downarrow}$ is favored and the system turns superconductive but, on the other hand, if $U > 0$ then the correlation $(\rho^{\rm ee}_{i\uparrow j\uparrow} - \rho^{\rm ee}_{i\downarrow j\downarrow})/2$ is instead favored and the system turns ferromagnetic. However, although $U < 0$ is a prerequisite for a non-zero superconductive correlation it alone does not guarantee it; this can be, for example, due to a strong enough Zeeman effect that forcefully magnetizes the system.

Schematic models of the hybrid and intrinsic self-consistent Majorana nanowires are shown in Fig.(1)(b). The intrinsic model is trivially described by Eq.(3), directly incorporating all terms. The hybrid system, however, comprises a depleted semiconducting (SM) nanowire (bottom in gray) and a doped superconducting (SC) nanowire (top in blue); thus $\mu_{\rm SC} \gg \mu_{\rm SM}$. Both nanowires are subject to the same Zeeman field but only the SM exhibits spin-orbit coupling. Crucially, only the SC nanowire contains the self-consistent Hubbard term $U_{\rm SC}$ (therewith $U_{\rm SM} = 0$), thus serving as the sole source of pairing within the system. Nevertheless, due to the proximity of the nanowires, there is a possibility for Cooper pairs to tunnel through the SC to the SM, effectively inducing superconductivity in the SM. This proximity effect is modeled by a hopping with intensity $\tau < t$.

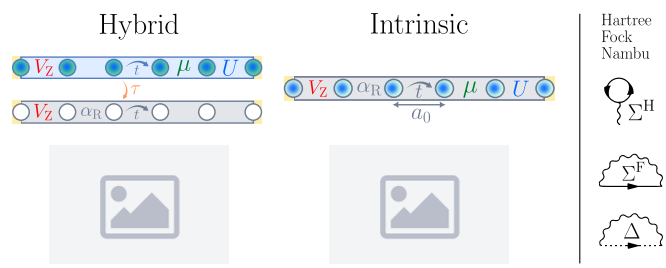

Figure 1. Models with DOS and self-energies diagrams

For both cases, the real-space lattice spacing was set to $a = 10$nm, the SOC Rashba strength to $\alpha = 40$meV and the finite nanowire lengths to $L = 2000$nm as to avoid to much overlap and hybridization of the Majorana's wavefunctions, localized at the edges of the nanowire but exponentially decaying into it's center. Moreover, we determined that a Hubbard strength of $U = -8$meV is optimal for achieving the largest possible gap rebound possible. Likewise, the optimal set of parameters for the hybrid case is $U_{\rm SC} = -30$meV, $\mu_{\rm SC} = 5$meV and $\tau = 0.8t$. These parameters are experimental feasible when considering an indium arsenide (InAs) semiconductor ($m^* = 0.023m_e$) and an aluminum (Al) superconductor, apart from the exceptionally low Fermi energy in the aluminum, typically $\mu_{\rm Al} \sim 11.6$eV. Nevertheless, this blunt variation in doping is crucial for facilitating interactions between

the SC to the SM, as the parallel momentum must be conserved across the interface. Subsequently, since we do not account for disorder (despite Al being well within the dirty limit), that tremendous mismatch in the Al/InAs Fermi surfaces would effectively nullify the proximity effect. why exactly?

As a pertinent aside, despite having the capability to accommodate multiple transverse modes, we chose to simplify our analysis by focusing exclusively on single-mode nanowires. This condition is ensured by maintaining a small radius R relative to the Fermi wavelength, $R < k_F^{-1}$. In the case of multi-mode nanowires, Majoranas could simultaneously emerge at the edges of different modes and destructively interfere (although this interference could be mitigated by a sufficiently strong spin-orbit coupling in the transverse direction).

III. RESULTS

We mainly focus the analysis on how the spectral gap behaves as a function of Zeeman V_Z and chemical potential μ, more particularly on the closing and reopening of the symmetry gap, commenting on the additional subtlety of self-consistence. For both hybrid and intrinsic models, we start by examining the cases of infinite length nanowires since these lay out much of the necessary groundwork without muddying it with finite length effects. Foremost, we identified the appropriate set of parameters, $\{a_0, \alpha, U\}$ for intrinsic and $\{a_0, \alpha_{\rm SM}, U_{\rm SC}, \mu_{\rm SC}, \tau\}$ for hybrid, by analyzing the band spectrum such as to ensure the visibility of the gap rebound and only then proceeded to the phase diagrams V_Z vs μ show in Fig.(3) and Fig.(2). In the hybrid analysis, to ensure that the shallow mode (and subsequently the gap) refers unequivocally to the SM, one must offset it's band structure by $\mu_{\rm shallow} = 2\tau\cos(\pi/3)$. why exactly?

Concerning the self-consistent routine, for the finite case we straightforwardly perform a full spectrum diagonalization $\rho = -i(\epsilon + i\eta - H)^{-1}$ [???] while for the infinite case we perform a Schur's factorization [???] where the rDM is calculated through $\rho_{dn} = \frac{1}{2\pi}\int_{-\pi}^{+\pi} d\phi f(H(\phi) - \mu I)e^{idn\phi}$ with ϕ the Bloch phase and f the Fermi function at zero temperature. As for the fixed-point computations, we utilized the so called Anderson acceleration/mixing method [???].

Finally, we present in Fig.(3) and Fig.(2) the self-consistent results for the intrinsic and hybrid nanowires, respectively. We begin by examining the hybrid case, initially focusing on the infinite nanowire and then moving on to the finite scenario. Subsequently, we address the more nuanced intrinsic case emphasizing on the differences between those models and the Oreg-Lutchyn minimal model.

In Fig.(2)(a), we present the spectral gap phase diagram $V_{\rm Z}$ versus μ, denoted as Ω. The diagram reveals two distinct regions demarcated by the Pffafian, the signature topological invariant also known as Majorana number $\mathcal{M}$; $\mathcal{M} = +1$ if it's in the trivial phase and $\mathcal{M} = -1$ if in the topological phase. For general non-interacting systems, the fermionic parity P of the ground state of a Hamiltonian H can be expressed as form $P(H) = \mathrm{sgn}(\mathrm{Pf}(A))$, where Pf denotes the Pfaffian of a skew-symmetric matrix A, given by $(\mathrm{Pf}(A))^2 = \det(A)$. The topological region takes the form of a hyperboloid, as predicted by the minimal model following the relation $(V_{\rm Z}/\Delta_0)^2 - (\mu/\Delta_0)^2 = 1$ under the assumption of a homogeneous pairing potential. Note, however, that the hyperboloid in not centered at $\mu = 0$, as is the case for the minimal model, but is instead rigidly shifted by the self-consistently induced doping. Is is relevant to note that, due to the particle-hole symmetries inherited from the Nambu formalism, there exists a holeonic phase-transition hyperboloid that is mirror-symmetric to the one show, appearing at a larger $\mu_{\rm SM}$.

Let us focus on the $V_{\rm Z}$ linecut at $\mu_{\rm SM} \approx -7.8$meV, shown in bright red in Fig.(2)(c). This specific value of $\mu_{\rm SM}$, denoted for reference as $\bar{\mu}_{\rm SM}$, corresponds to the trivial-topological crossing at the edge of the phase transition hyperboloid, highlighted by the dashed bright red line in Fig.(2)(a). At $V_{\rm Z} = 0$ the system is in its trivial phase, with the gap decreasing linearly as $V_{\rm Z}$ increases, reaching zero precisely at the white boundary, marked by an abrupt change in the Pfaffian. This behavior is expected since the gap is always maintained due to pairing, but as $V_{\rm Z}$ increases, the system becomes more strongly magnetically induced, effectively suppressing s-wave superconductivity within the trivial region. After this, the gap unexpectedly reopens, up to the maximum of $\Omega \approx 0.3$meV, signaling the topological phrase transition and indicating that the system now effectively behaves as a p-wave superconductor. Following the rebound, the gap exponentially decays with $V_{\rm Z}$ until it asymptotically approaches zero. These results closely mirror those predicted by the Oreg-Lutchyn minimal model; however, instead of a exponential decay to zero, we observe a smoother asymptotically decay toward the fixed homogeneous pairing value Δ_0. right? This dissimilarity arises from the self-consistent treatment, which, under a strong enough Zeeman field, magnetism will always dominate over pairing.

We now redirect our focus over to the $\bar{\mu}_{\rm SM}$ linecut at $V_{\rm Z} = 0$meV, shown in bright green in Fig.(2)(b). I the absence of a Zeeman field and with $U_{\rm SC} < 0$, the spectral gap can only arise due to the SC induced pairing or direct doping. Within the lower region of Fig.(2)(a), the SM is effectively depleted, behaving as an insulation, with the gap growing as $\mu_{\rm SM}$ decreases. However, as $\mu_{\rm SM}$ nears $\bar{\mu}_{\rm SM}$, the gap must be supported exclusively by the pairing, leading to a very gradual decay as $\mu_{\rm SM}$ increases. right?

Additionally, the same $\bar{\mu}_{\rm SM}$ and $V_{\rm Z} = 0$ linecuts where recalculated but without SOC (thereby having set the Rashba strength α to zero before self-consistency), being depicted in soft green and soft red in Fig.(2)(b) and (c), respectively. Comparing both linecuts, with and without

SOC, clearly demonstrates the critical role of SOC in the topological phase transition. Specifically, as shown in Fig.(2)(c), without SOC the gap fails to reopen indicating the absence of a phase transition. The SOC role on the μ linecut is not so relevant, indicating just a rigid shift on the phase transition hyperboloid and an inward lump at it's center I assume this from the intrinsic results since it seems that the $\alpha = 0$ results did not converge. I will check this once I have the actual results.

Furthermore, we compare the $\bar{\mu}_{\rm SM}$ and $V_{\rm Z} = 0$ linecuts without SOC to the analytical gap equation, generalized to account for Zeeman but not SOC, derived in appendix B:

$$\Delta_0 = 2\epsilon_0 \exp\left(-\sqrt{\frac{\pi^2}{2}\frac{\hbar^2}{m_e e}}\sqrt{\frac{\mu}{m_0}}\frac{1}{|U|a_0}\right), \qquad (4)$$

It is important to note that Eq.(4) assumes $\mu \gg \Delta$. See that the actual spectral gap Ω (show in dashed black) does not always correspond to the pairing potential Δ_0. Generally it reads $\Omega = \max(0, |\Delta_0| - |V_{\rm Z}|)$ since for $|V_{\rm Z}| > \Delta_0$ the gap has to close although the pairing remains finite. Moreover, see that one always needs $\mu > 0$ or else the pairing becomes non-real. In the particular case $V_{\rm Z} = 0$ with $\mu < 0$ the gap must correspond instead to $\Omega = -\mu$. Comparing Eq.(4) to the self-consistent results without SOC in Fig.(2)(b) we see that...

Moving on to the finite scenario, we shift our focus to the analysis of Fig.(2)(d) and (e). As one transitions from the trivial to the topological phase, characterized by the closing and reopening of the gap, two zero-energy modes emerge, corresponding to Majorana bound states (MBs) localized at each edge of the finite nanowire, and exponentially decaying into it's center. These states can be identifies by the wavy patterns observed within the topological region, arising from the overlap and interface of the Majoranas' wavefunctions. This oscillatory behavior is further highlighted in Fig.(2)(e) where the energy spectrum ϵ is plotted as a function of $V_{\rm Z}$ at fixed $\bar{\mu}_{\rm SM}$. For a given $V_{\rm Z}$ within the topological region, the density of states (DOS) below the topological minigap of the quasicontinuum would exhibit two peaks near zero energy or just the single one at twice the height at the cross. Since Majoranas decay exponentially fast from their edge, for sufficiently long wires, the interference becomes minimal, and the energy spectrum reflects that with a flat line precisely at zero energy.

To highlight once more the crucial role of SOC, we plot in the background of Fig.(2)(e), and in light gray, the energy spectrum ϵ having set $\alpha = 0$ before self-consistency. Without SOC, there is neither a helical gap nor canting of spin, which altogether disables the formation of s-wave pairing, thereby effectively suppressing superconductivity within the system. This suppression is reflected in the energy spectrum, which remains closed after the point where the phase transition would occur. Additionally, it is important to note that a strong SOC exponentially suppresses Majorana oscillations.

All the results pretending to the finite hybrid nanowire also closely mirror the ones predicted by the Oreg-Lutchyn minimal model, but without the unrealistic imposition of a fixed homogeneous pairing potential.

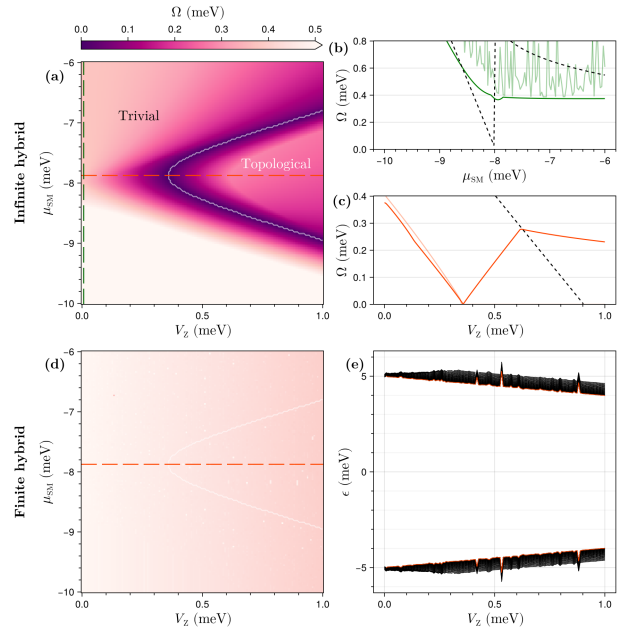

Figure 2. **Self-consistent SC-SM hybrid Majorana nanowire results. (a)** Zeeman strength $V_{\rm Z}$ versus the SM chemical potential $\mu_{\rm SM}$ spectral gap Ω phase diagram. The white boundary line denotes the abrupt change in the Majorana number and marks the trivial-topological phase transition. **(b)** Spectral gap $\Omega(V_{\rm Z} = 0)$ vs SM chemical potential $\mu_{\rm SM}$. The self-consistent results with SOC are shown in dark green, the ones without in soft green The parameters used are: lattice spacing $a = 10$nm, finite nanowire length $L = 2000$nm, SOC Rashba strength $\alpha = 40$meV, SC Hubbard potential strength $U_{\rm SC} = -30$meV, SC Hubbard chemical potential $\mu_{\rm SC} = 5$meV and SC-SM hopping coupling $\tau = 0.8t$ with effective mass $m^* = 0.023m_e$.

For the most part the intrinsic results shown in Fig.(3)

Comparing Eq.(4) to the self-consistent results without SOC in Fig.(3)(c) reveals a strong correspondence; the linear regression fits perfectly and the exponential decaying curves has the analytics converging towards the numerics as expected. The remain deviation could be justified due to assuming $\mu \gg \Delta$ in the analytics. why? Concerning Fig.(3)(b), the analytics-numerics correspondence is to be expected due to it's simplicity. lets see

- **Written part**

1. **insulator part only on intrinsic (?)**
2. **intrinsic decay more rapid**
3. **analytics for $V_{\rm Z}$ linecuts**
4. **higher peak on hybrid without needing HUGE $V_{\rm Z}$**

5. **captions**

- **Results**

1. **infinite hybrid**

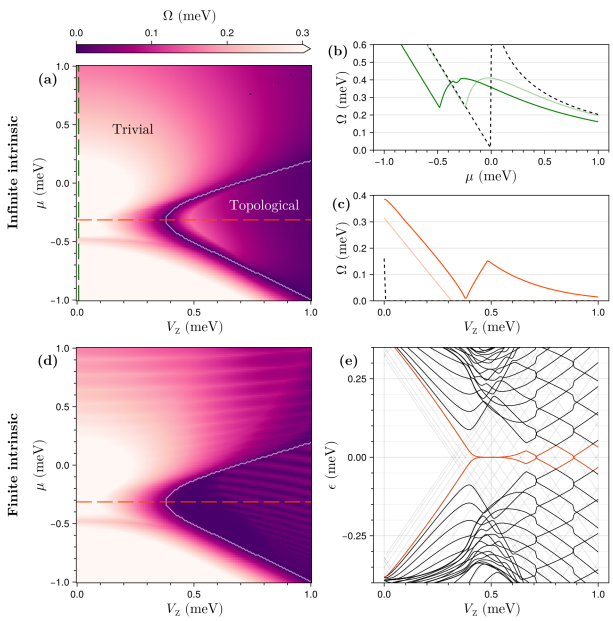

Figure 3. Intrinsic self-consistent Hartree-Fock-Bogoliubov nanowire

IV. CONCLUSIONS

ACKNOWLEDGMENTS

We are deeply grateful to our colleague Rui E. Silva who provided useful insight and expertise that greatly assisted the research of this paper.

APPENDIX: SELF-CONSISTENT SUPERCONDUCTIVITY

Appendix A: Hartree-Fock-Bogoliubov mean-field theory

Consider a generic tight-binding quartic Hamiltonian,

$$\hat{H} = \sum_{\alpha\beta} h_{\alpha\beta} c^\dagger_\alpha c_\beta + \frac{1}{2} \sum_{\alpha\beta\gamma\delta} V^{\alpha\beta}_{\gamma\delta} c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta, \tag{5}$$

with $c^\dagger_\alpha (c_\alpha)$ fermionic creation (annihilation) operators of site and spin degree of freedom $\alpha = (i, \sigma_i)$, and $V^{\alpha\beta}_{\gamma\delta}$ a spinless electrostatic potential tensor of symmetries $V^{\alpha\beta}_{\gamma\delta} = V^{\beta\alpha}_{\delta\gamma}$ and $(V^{\alpha\beta}_{\gamma\delta})^* = V^{\gamma\delta}_{\alpha\beta}$ following directly from $V(\mathbf{r} - \mathbf{r}') = V(\mathbf{r}' - \mathbf{r})$. We fabricate a doubling of the degrees of freedom by introducing the Nambu-spinor $\check{c}_a = (c_a \; c^\dagger_a)$, constraining a particle-hole symmetry. In this Nambu representation the system's reduced density matrix (rDM) corresponds to the tensor product $\rho_{ab} = \langle \check{c}^\dagger_b \otimes \check{c}_a \rangle$, reading explicitly as

$$\rho_{ab} = \begin{pmatrix} \langle c^\dagger_b c_a \rangle & \langle c_b c_a \rangle \\ \langle c^\dagger_b c^\dagger_a \rangle & \langle c_b c^\dagger_a \rangle \end{pmatrix} = \begin{pmatrix} \rho^{\mathrm{ee}}_{ab} & \rho^{\mathrm{eh}}_{ab} \\ \rho^{\mathrm{he}}_{ab} & \rho^{\mathrm{hh}}_{ab} \end{pmatrix}, \tag{6}$$

with it's terms relating to each other directly from the fermionic anti-commutator properties.

Performing a mean-field Hartree-Fock decoupling as follows $\langle c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta \rangle \approx \rho^{\mathrm{he}}_{\beta\alpha} \rho^{\mathrm{eh}}_{\delta\gamma} - \rho^{\mathrm{ee}}_{\gamma\alpha} \rho^{\mathrm{ee}}_{\delta\beta} + \rho^{\mathrm{ee}}_{\delta\alpha} \rho^{\mathrm{ee}}_{\gamma\beta}$, the Nambu rDM equation of motion can be shown to read simply as $i\hbar \dot{\rho}_{ab} = \left[H^{\mathrm{BdG}}_{ab}, \rho_{ab} \right]$ with

$$H^{\mathrm{BdG}}_{ab} = \begin{pmatrix} h^{\mathrm{HF}}_{ab} & \Sigma^{\Delta}_{ab} \\ \left(\Sigma^{\Delta}_{ba}\right)^* & -h^{\mathrm{HF}}_{ba} \end{pmatrix} \tag{7}$$

the effective Bogoliubov-de Gennes (BdG) Hamiltonian, $h_{\mathrm{HF}}(\rho) = h + \mathbf{\Sigma}_H(\rho) + \mathbf{\Sigma}_F(\rho)$ the Hartree-Fock Hamiltonian, having defined the Hartree, Fock and pairing self-energies as $\Sigma^H_{ab} = \sum_{\gamma\delta} V^{\delta a}_{b\gamma} \rho^{\mathrm{ee}}_{\gamma\delta}$, $\Sigma^F_{ab} = -\sum_{\gamma\delta} V^{a\delta}_{b\gamma} \rho^{\mathrm{ee}}_{\gamma\delta}$ and $\Sigma^\Delta_{ab} = \sum_{\gamma\delta} V^{ab}_{\delta\gamma} \rho^{\mathrm{eh}}_{\gamma\delta}$. Note the pairing symmetry $\Sigma^\Delta_{ab} = -\Sigma^\Delta_{ba}$.

We build upon these analytical results by further considering the approximation of ultra-localized orbitals $V^{ij}_{kl} \approx V^{ij} \delta_{il} \delta_{jk}$ with a spinless potential $V^{i\sigma_i j\sigma_j} = V^{ij}$, followed by the exclusion of all offsite term contribution through $V^{ij} = U \delta_{ij}$. Defining a new BdG symmetric object

$$\tilde{\boldsymbol{\rho}}_{ij} = \begin{pmatrix} \boldsymbol{\rho}^{\mathrm{ee}}_{ij} & -\boldsymbol{\rho}^{\mathrm{eh}}_{ij} \\ -\left(\boldsymbol{\rho}^{\mathrm{eh}}_{ji}\right)^* & -\boldsymbol{\rho}^{\mathrm{ee}}_{ji} \end{pmatrix}, \tag{8}$$

related to the Nambu rDM $\boldsymbol{\rho}_{ij} = \langle \check{\mathbf{c}}^\dagger_j \otimes \check{\mathbf{c}}_i \rangle$ with $\check{\mathbf{c}}_i = (c_{i\uparrow} \; c_{i\downarrow} \; c^\dagger_{i\uparrow} \; c^\dagger_{i\downarrow})$ and the underlying spin structure concealed within the bold notation allows one to write the total onsite self-energy of the system in the compact form

$$\mathbf{\Sigma}_{ij} = \frac{1}{2} V^{i\ell} \mathrm{Tr} \left(\left[\tau_z \otimes \sigma_0 \right] \tilde{\boldsymbol{\rho}}_{\ell\ell} \right) \left[\tau_z \otimes \sigma_0 \right] \delta_{ij} - V^{ij} \tilde{\boldsymbol{\rho}}_{ij}, \tag{9}$$

with both σ and τ being Pauli matrices operating in their respectively spin and electron-hole subspaces and the Kronecker deltas δ_{ij} denoting on-site terms and $\delta^{\pm}_{ij\pm1}$ the 1st neighbors hoppings (front and back). Note the repeated dummy Einstein site index ℓ concealing the hidden sum.

Appendix B: Analytical gap equation

The main objective of this section is to derive the typical textbook BCS gap equation for an infinite system without Zeeman or SOC [??] from our Hartree-Fock-Bogoliubov approach. We generalize the result for a Zeeman field, accounting not only for the self-consistent pairing but also the self-consistently induced magnetization

and doping. We will not, however, account for SOC effects.

Foremost, let us reparametrize the self-energy of Eq.(2) in terms of the self-consistently induced magnetization $\delta V_{\rm Z}$, doping $\delta\mu$ and pairing Δ (assumed to be real) as $\mathbf{\Sigma}_{ii} = (\delta V_{\rm Z}\,[\tau_z \otimes \sigma_z] - \delta\mu\,[\tau_z \otimes \sigma_0]) - \Delta\,[\tau_y \otimes \sigma_y]$. Comparing this rewritten form with the original one, we can identifying the off-diagonal block $\Sigma^{\rm he}_{ii}$ as being $-i\Delta\sigma_y = U\rho^{\rm eh}_{i\uparrow i\downarrow}$, noting that $\rho^{\rm eh}_{i\uparrow i\downarrow} = -\rho^{\rm eh}_{i\downarrow i\uparrow}$.

In another context, consider the Hamiltonian (no SOC) $H_0(k) = \sum_s c^\dagger_{ks}\left[(\epsilon_k - \mu)\delta_{ss'} + V_Z\sigma_z\right]c_{ks'}$ with s spin and k momentum. For the Hamiltonian at hand we have $\rho^{\rm eh}_{k\uparrow\downarrow} = \frac{1}{N}\sum_k\left[u^*_{k+}v_{k+}f(\tilde{E}_{k+}) + u^*_{k-}v_{k-}f(\tilde{E}_{k-})\right]$ where $\tilde{E}_{k\pm} = \tilde{V}_{\rm Z} \pm \tilde{E}_k$ with $\tilde{E}_k = \sqrt{\Delta^2 + \tilde{\epsilon}^2_k}$ (no $\pm$ subindex) are its eigenvalues and $(u_{k\pm},\, v_{k\pm})$ the (normalized) eigenvectors of the $\uparrow\downarrow$ block; $\tilde{\epsilon}_k = \epsilon_k - \tilde{\mu}$, $\tilde{\mu} = \mu + \delta\mu$ and $\tilde{V}_{\rm Z} = V_{\rm Z} + \delta V_{\rm Z}$ already account for the U-induced shift and f is the Fermi-Dirac distribution. It is critical to realized that is in this generalization we have additional, implicit, self-consistent condition for both $\delta V_{\rm Z}$ and $\delta\mu$. These are obtained in a similar fashion to the pairing term but considering instead the diagonal elements of $\mathbf{\Sigma}_{ii}$. The self-consistent system of three equation for pairing Δ, induced magnetization $\delta V_{\rm Z}$ and doping $\delta\mu$ read as

$$\Delta = -\,U\sum_k \frac{\Delta}{4\tilde{E}_k}\left[\tanh\left(\frac{\tilde{V}_{\rm Z} + \tilde{E}_k}{2k_BT}\right) - \tanh\left(\frac{\tilde{V}_{\rm Z} - \tilde{E}_k}{2k_BT}\right)\right] \tag{10}$$

$$\delta V_{\rm Z} = -\,\frac{U}{2N}\sum_k \frac{\sinh\left(\frac{\tilde{V}_Z}{k_BT}\right)}{\cosh\left(\frac{\tilde{V}_Z}{k_BT}\right) + \cosh\left(\frac{\tilde{E}_k - \tilde{V}_{\rm Z}}{k_BT}\right)} \tag{11}$$

$$\delta\mu = \frac{U}{4} + \frac{U}{4N}\sum_k \frac{\tilde{\epsilon}_k}{2\tilde{E}_k}\left[\tanh\left(\frac{\tilde{V}_{\rm Z} + \tilde{E}_k}{2k_BT}\right) - \tanh\left(\frac{\tilde{V}_{\rm Z} - \tilde{E}_k}{2k_BT}\right)\right] \tag{12}$$

See that, if one ignores in Eq.(10) Zeeman all together, $V_{\rm Z}$ and $\delta V_{\rm Z}$, but also the self-consistently doping $\delta\mu$ it reduces to the traditional gap equation [??] as intended. Note that this has a nonzero solution only if the interaction is attractive i.e $U < 0$. Although these analytical expressions can be obtained, the generalized analytical solution of the self-consistent system of equation is neither trivial nor worth the trouble for our purposes. We can, however, solve the conventional gap equation in the zero-temperature, small-gap limit, which offers a sufficiently accurate analytical comparison to our numerical results. Thereby, we approximate the dispersion to $\epsilon_k = \pm v_{\rm F}(k \mp k_{\rm F})$ with $v_{\rm F}$ and $k_{\rm F}$ the Fermi velocity and wavevector respectively, and mutate the $\frac{1}{N}\sum_k$ sum into an integral over ϵ_k, introducing also a fitted phenomenological cutoff ϵ_0. In the limit where $\Delta \ll \epsilon_0$ and for $U < 0$ the gap equation simplifies to

$$\Delta = 2\epsilon_0 \exp\left(-\sqrt{\frac{\pi^2}{2}\frac{\hbar^2}{m_e e}}\sqrt{\frac{\mu}{m_0}}\frac{1}{|U|a_0}\right). \tag{13}$$

Even though we neglected Zeeman this solution is still valid for finite $|V_{\rm Z}| \leq \Delta$ since the tanh factors remain equal to 1 throughout the integration interval. Nevertheless, the generalization of Eq.(13) for $|V_{\rm Z}| > \Delta$ can be found from Eq.(10) within the same regime of approximation and expressed in terms of the previous $\Delta_0 \equiv \Delta(V_{\rm Z} = 0)$ solution as

$$\Delta = \frac{1}{2\epsilon_0}\sqrt{\Delta_0\left(2\,|V_{\rm Z}| - \Delta_0\right)}. \tag{14}$$

It is important to note that the actual spectral gap Ω does not always correspond to the pairing potential Δ_0. Generally it reads $\Omega = \max(0, |\Delta_0| - |V_{\rm Z}|)$ since for $|V_{\rm Z}| > \Delta_0$ the gap has to close although the pairing remains finite. Moreover, see that one always needs $\mu > 0$ or else the pairing becomes non-real. In the particular case $V_{\rm Z} = 0$ with $\mu < 0$ the gap must correspond instead to $\Omega = -\mu$.

[0.0] Lutchyn, Roman M., Jay D. Sau, and S. Das Sarma. "Majorana fermions and a topological phase transition in semiconductor-superconductor heterostructures." Physi-

cal review letters 105.7 (2010): 077001.

[0.1] Oreg, Yuval, Gil Refael, and Felix Von Oppen. "Helical liquids and Majorana bound states in quantum wires." Physical review letters 105.17 (2010): 177002.

[0.2] Chamon, C., et al. "Quantizing Majorana fermions in a superconductor." Physical Review B 81.22 (2010): 224515

[0.3] Prada, Elsa, et al. "From Andreev to Majorana bound states in hybrid superconductor–semiconductor nanowires." Nature Reviews Physics 2.10 (2020): 575-594..

[0.4] Avila, J., et al. "Superconducting islands with semiconductor-nanowire-based topological Josephson junctions." arXiv preprint arXiv:2003.02852 (2020).

[0.5] Sau, Jay D., et al. "Generic new platform for topological quantum computation using semiconductor heterostructures." Physical review letters 104.4 (2010): 040502.

[0.6] Fu, Liang, and Charles L. Kane. "Superconducting proximity effect and Majorana fermions at the surface of a topological insulator." Physical review letters 100.9 (2008): 096407.

[1.1] Nambu, Yoichiro. "Quasi-particles and gauge invariance in the theory of superconductivity." Physical Review 117.3 (1960): 648.